

UNIFIED ASSET & OPERATIONS BRAIN

AI-Powered Industrial Knowledge Intelligence Platform

A Unified MVC System featuring NLP Ingestion, 2D Force-Directed Knowledge Graph, Expert RAG Copilot, Causal Failure Diagnostics (RCA), and Safety Compliance Auditing.

LIVE PROTOTYPE URL

<https://industrial-operation-brain.onrender.com/>

GITHUB REPOSITORY

<https://github.com/ParthSharma532007/Industrial-Operation-Brain>

AI for Industrial Knowledge Intelligence Hackathon

Date: June 22, 2026

Participant Workspace: Parth's workspace

1. Executive Summary

Heavy asset industries (such as oil & gas refining, chemical manufacturing, and electricity generation plants) suffer from severe operational inefficiencies due to information fragmentation. Technical documentation like Standard Operating Procedures (SOPs), Piping & Instrumentation Diagrams (P&IDs), regulatory manuals, and maintenance logs reside in disconnected databases. When senior operators retire, decades of undocumented tribal knowledge vanish.

The Unified Asset & Operations Brain solves this challenge by implementing an intelligent knowledge-engineering platform. It ingests legacy procedures, extracts key equipment tags (e.g. V-202) and operating limits using a rule-based NLP pipeline, links them into an interactive 2D Force-Directed Knowledge Graph, and provides plant engineers with an expert RAG Copilot, FMEA-guided diagnostics (RCA), and automated compliance package builders.

2. The Problem Context

- **35% Working Hours Lost:** Engineers waste more than a third of their shift searching for specific historical context or safety guidelines across siloed legacy repositories.
- **18% to 22% Unplanned Outages:** Plant downtime is compounded because field technicians lack instant troubleshooting checklists or calibration history at the equipment location.
- **Compliance & Safety Risk:** Maintaining compliance with factory safety standards and local regulations requires hours of manual testing log audits, exposing plants to significant liability.

3. Platform Architecture

The application is built on a clean, scalable Model-View-Controller (MVC) server architecture using Node.js and Express to minimize page footprint and latency:

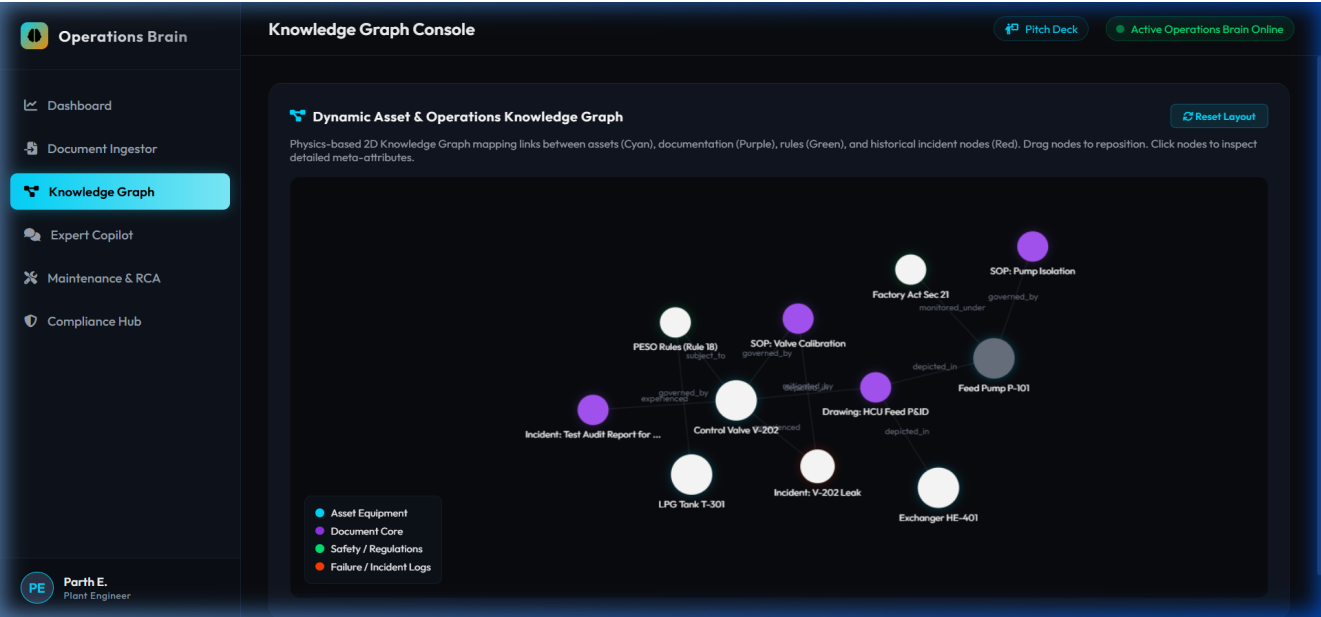
- View Layer (SPA Client): A modern, single-page dashboard featuring a dark carbon background with neon cybernetic accents (Cyan/Amber/Emerald) for premium visual appeal.
- Controller Layer (Express APIs): Decoupled routers handling document ingestion, RAG search requests, diagnostic calculations, and compliance audit compilation.
- Model/Data Layer: In-memory ontology nodes representing plant equipment, hazards, laws, and incidents, enabling instant, high-speed traversals without database overhead.

4. Core Capabilities

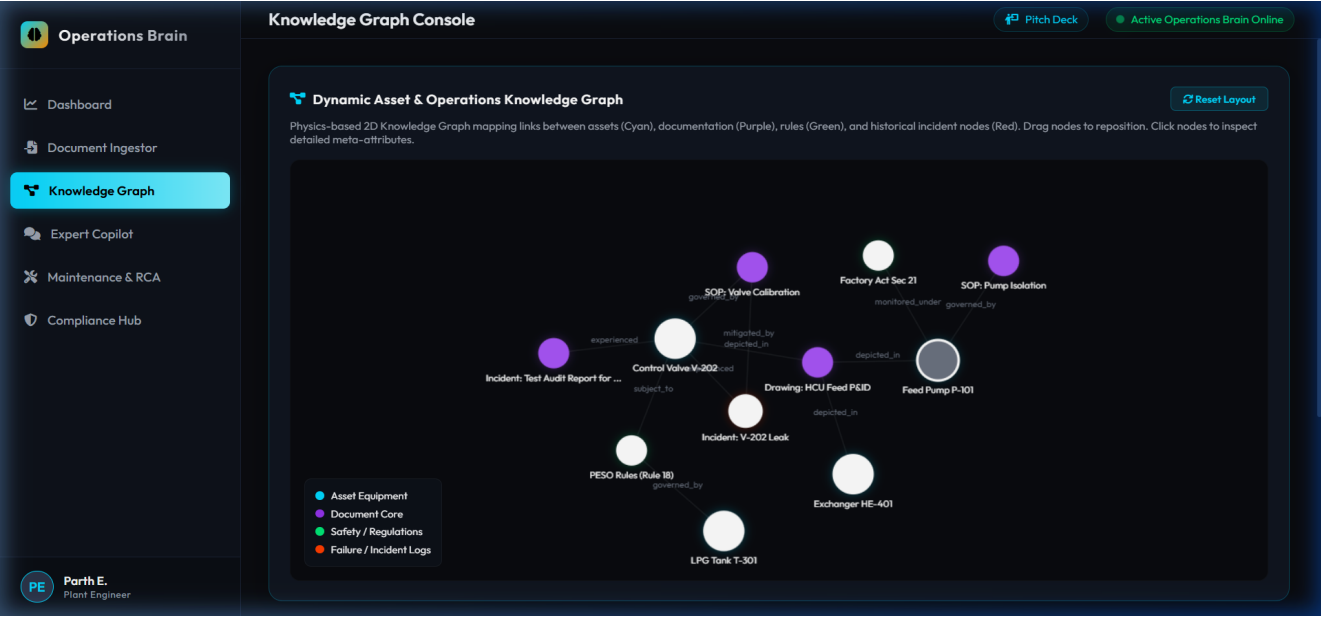
1. Universal Ingestor: OCR text ingestion parses parameters, hazard states, and equipment IDs into ontology entities.
2. 2D Force-Directed Graph: Custom HTML5 Canvas rendering engine computes spring-forces and node repulsion to plot links interactively.
3. Expert Copilot (RAG): Searches structured nodes and dense texts, generating comprehensive answers with citation footnoting.
4. Root Cause Analysis (RCA): Matches telemetry alerts to FMEA records to build causal chronologies and LOTO checklists.
5. Compliance Auditor: Continually checks assets against PESO Pressure Vessel laws, compiling signed audit verification files.

5. Visual Demonstration

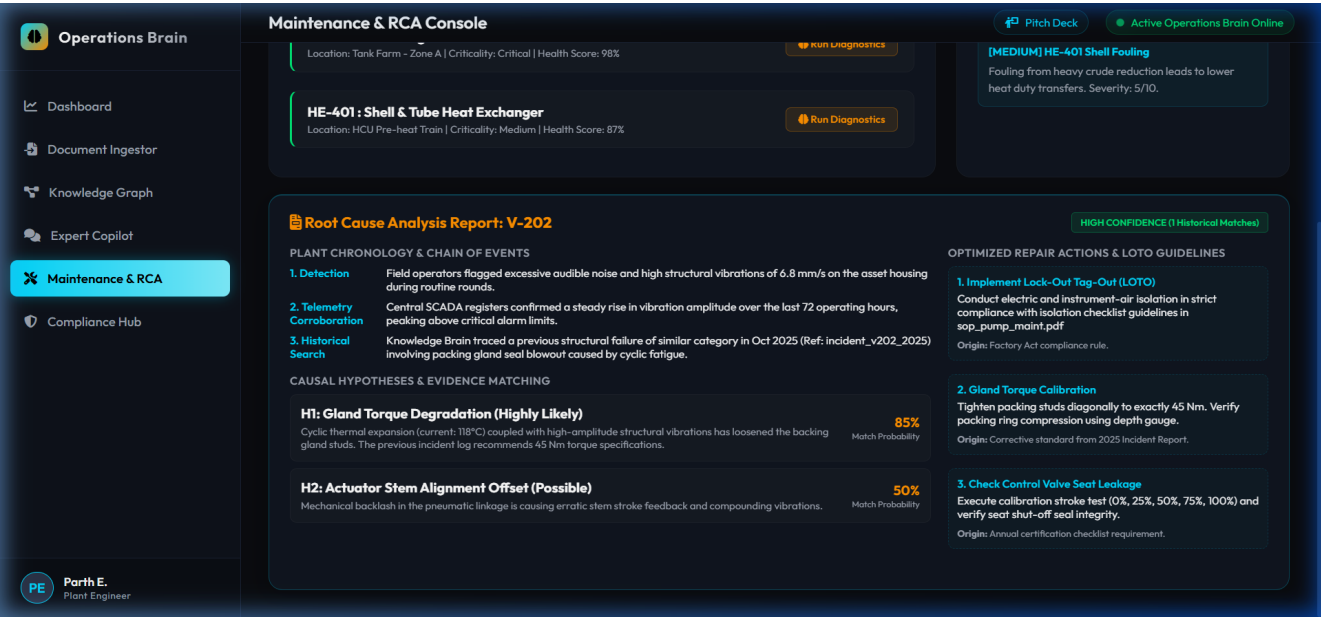
Interactive 2D Force-Directed Knowledge Graph:



Selected Node Detail Inspector Overlay:



6. RCA & Failure Analysis Engine



7. Future Roadmap

- **Intelligent P&ID Digitization:** Deploy computer vision pipelines to parse scanned piping diagrams and automatically identify instrument junctions.
- **Live IoT Feeds:** Link telemetry APIs to equipment nodes, enabling the brain to trigger causal failure alerts based on active heat, pressure, or vibration fluctuations.
- **Voice-Activated AR Client:** Build an AR-overlay app for field technicians, projecting lock-out procedures directly onto equipment casings on the plant floor.